

Executive Summary

Introduction

The Search for Extraterrestrial Intelligence (SETI) is a series of projects conducted and administered by different countries with a common goal: *to search the sky for electromagnetic evidence of the existence of other intelligent life forms beyond our solar system.*

Before 1992, SETI searches were conducted on a limited scale by NASA, the Planetary Society, some universities, a few astronomical organizations and individual investigators. The pace of the search was markedly accelerated in October, 1992, when NASA began a detailed survey of selected stars and a full-sky survey. The NASA project, known as the High Resolution Microwave Survey (HRMS), uses radio telescopes outfitted with specially-developed high speed computer systems. It eclipses all previous searches and reinforces the need to seriously address a significant question related to the outcome

What happens if we find intelligent extraterrestrial life?

This report is the product of a series of Workshops convened by NASA, with the support of the SETI Institute, to examine the implications for human society of such a discovery.

The chapters of this document describe NASA's Microwave Survey. They examine the idea of extraterrestrial life through recent history, analyze ways in which people and groups might react to a detection of extrater-

restrial intelligence (ETI) or to a long search with no detection, and assess possible reactions to a detection by political and governmental entities. A final chapter examines the roles of educational institutions, news media, and entertainment media as informers of the public in the event of a detection.

The chapters seek to identify studies and actions that ensure that the public and various political entities and organizations can be better acquainted with SETI and more likely to respond to an announcement of a detection in ways that are orderly, positive, and beneficial. Each chapter concludes with a list of recommendations.

The highlights of the workshops findings are summarized in the following.

Principal Findings and Recommendations

Chapter 1. The NASA Observing Program — A High Resolution Microwave Survey

Detection of an ETI signal would spark intense widespread interest that would in turn prompt new technological advances and highlight a need for organized assessment of the discovery. The discovery would contribute to philosophical discussions and perspectives. Non-detection after a long interval would likewise suggest insights about humanity's situation in the universe, perhaps reflect-

ing an apparent scarcity, uniqueness, and loneliness of intelligent communicating beings.

Chapter 2. History and SETI

For several centuries people have accepted the possibility that other worlds house intelligent beings. Whereas many have been relatively indifferent to this idea, others have sorted themselves into "millennial" and "catastrophist" camps, respectively espousing positive or negative views of the implications of ETI existence. In recent years both views have continued to attract adherents, now using modern terminology to argue for or against efforts to detect ETI's. Modern arguments about ETI's focus on the Drake equation, which gives highly varying assessments of the abundance or scarcity of ETI's in our galaxy.

Views of SETI are confused by public perceptions that physical contact between human and ETI cultures must inevitably be involved. Cultural contacts in history have had unfortunate outcomes for many people, prompting negative analogies with SETI. In contrast to prevailing impressions, however, some cultural contacts have resulted in outcomes in which all parties benefited. In particular, entry of Arab knowledge to medieval Europe, with consequent flowering of European scholarship and subsequent discovery, may be analogous to the Microwave Survey's potential to provide advanced new information without physical contact. There are no historic events that are exactly analogous to an ETI detection. This mandates caution in seeking insights into likely human reactions to such a detection from partially analogous historic encounters and events.

Recommendations (Paraphrased; see Chapter 2 for the full text.)

1. NASA should examine events that are partially analogous to an ETI detection, drawn from earlier human experience, but should also recognize that these provide only approximate guides for insights about SETI and not precise predictors of the future.
2. Study should be focused on historic analogies based on the transmission of ideas within and between cultures, rather than on analogies based on physical encounters between cultures.
3. NASA's educational programs should place SETI within the historical context of humankind's effort to comprehend its place in the universe and to understand the nature and possibility of other intelligent life.

Chapter 3. SETI and Human Behavior: Human Responses to an ETI Signal Detection

The factors that influence human responses include attributes of individuals (for example, gender, race or ethnic affiliation, religion, level of education) and attributes of the environmental backdrop (including nuances of languages used to describe events, whether outlooks are optimistic or pessimistic, belief systems, economic situations, cultural stores of images, and frames of reference). The reactions of individuals and groups depend upon interplays among factors, in ways that are not known or presently predictable.

Reactions to a detection (or non-detection) can range from indifference through mild positive or negative curiosity, through millennial

enthusiasm or catastrophist anxiety, to full scale pronia or paranoia. Most individual reactions to an announcement would include active expanded searches for additional information, with significant coalescences of like-minded individuals in support (or opposition) groups. A few reactions would probably be irrationally extreme or even violent.

Education is identified as a factor that correlates with positive attitudes toward SETI. Research is recommended that enables us to better understand factors that underlie various responses and to identify activities that increase the likelihood that responses will be well-informed and even positive. One promising fact-finding technique is to pose specific alternative ETI detection scenarios, then poll the public on likely responses to those scenarios.

Recommendations (Paraphrased; see Chapter 3 for the full text.)

1. Many cultures have traditions of depicting strangers, aliens, and ETI, and have different emotional responses to specific terms used within those traditions. Analyzing these terms (and perhaps coining new ones) will enable SETI participants to anticipate and pre-empt possible negative responses by adopting accurate language that is free of unexpected shadings of value.
2. There is need for research on popular public visions, perceptions, and images of ETI's, and on cultural conceptions of science and technology.
3. The preliminary approach that we have adopted in this report identifies separate variables that motivate behavior, but does not specify the ways in which interactions among the variables influence behavior.

Studies are needed to learn how these various elements interact.

4. We should try to identify and clarify the responses of various peoples to the Apollo and Viking missions (activities that were remotely analogous to NASA's HRMS research).
5. We should identify groups with unusual abilities to affect policy whose members might require better information about SETI. Can better understanding of SETI encourage their members to view these endeavors positively (or more positively)?
6. SETI researchers (or their designees) should identify individual and institutional responses that are viewed by the public as positive and achievable in the event of a signal detection, then participate in programs of education and information that will enable people to respond in those ways.
7. SETI researchers or their designees should make greater use of popular media, including movies, computer games, and popular music, to present SETI and ETI themes. These avenues could greatly increase public interest and exposure.
8. SETI researchers should identify, assist, and inform national and international organizations concerned with SETI-related issues.
9. SETI researchers should affiliate with "soft" science researchers through such bodies as the World Ethnographic Union. These relationships could bring relevant questions of behavior to the attention of those workers, and draw upon multiple disciplinary resources for understanding

those variables whose effects on human reactions are unclear.

10. The creation of a panel of expert behavioral scientists as a "reaction team" should be considered. Such a team should be available for advice and help in situations where information about an ETI signal seems to cause unusual disruption of normal patterns of life.
11. Means should be devised for including the perspectives and inputs of members of non-American, non-white societies in the future.
12. Great care must be taken that methodological assumptions are not made into metaphysical assumptions in describing SETI activities. We should not assume, as a matter of dogma, that ETI civilizations are eager to communicate with us, nor that all human beings are eager to communicate with them.

Chapter 4. SETI and Policy

Many organized entities are likely to try to use news of an ETI detection in ways that serve their purposes. These include promotion of claims of a "cover-up" of information about the detection, advocacy of increased aerospace research, maneuvers to manipulate the release of information about the find for political benefit, and efforts to control or suppress the release of information. Efforts to suppress or control information are unlikely to succeed, given the widespread SETI verification and data-sharing network and the likely ability of most nations to "tune in" an ETI source with modest equipment.

Government bodies will need to respond to news of an ETI detection. Most have no poli-

cy or mechanisms in place for responses, and could benefit from discussion now of possible actions to be taken in the event. One existing agreement on actions to be taken by SETI researchers in the event of a detection is widely endorsed. However, it has no legal status or government signatories at present.

An important question would be that of sending a reply to an ETI message or signal. That discussion would be guided by whether the ETI transmission was intended for us or was overheard in passing, whether it was information-rich or information-poor, "friendly" or "hostile" in tone, and by other considerations. Discussion should begin on whether or how international or national representative bodies should address this issue.

Because existing institutions, processes, and agreements generally neglect the possibility of an ETI detection, much remains to be done to prepare for policy responses to a detection. The work could be addressed in two broad categories: first begin the educational and informational processes necessary to prepare government bodies and international institutions most likely to be affected by an ETI detection, and second begin establishing procedures and mechanisms similar to those outlined in this chapter, to make humankind's responses to detection more effective.

Recommendations (Paraphrased; see Chapter 4 for the full text.)

1. NASA should encourage establishment of post-detection notification processes within and among concerned national governments and inter-governmental bodies identified by this report;
2. NASA should clarify its own policy and procedures for public dissemination of information about the detection;

3. In accordance with established diplomatic procedures, NASA should brief national governments and international organizations about the Microwave Survey project and the implications of detection;
4. NASA should conduct further analyses of the political consequences of detection;
5. In accordance with established diplomatic procedures, NASA should work with governments and international organizations to develop procedures within international organizations for responses to detection;
6. NASA should broaden international participation in SETI through training and technical assistance.

Chapter 5. SETI, Education, News, and Entertainment

Schools, libraries, museums and universities are able and likely to respond quickly with detailed accurate information on SETI, both in the short term if deluged with requests following a detection and in the long term via education programs. One weakness in this network is identified, a disproportionate availability of materials on UFO's and pseudoscientific aspects of extraterrestrial life in libraries alongside reliable SETI material. How educational institutions themselves would be changed by an ETI detection is not easy to predict. Studies of ways in which the somewhat analogous *Sputnik* launch or Percival Lowell's conjectures about intelligent life on Mars affected schools and texts may provide insights.

News media will be the source of information for most people during a detection event. Reliable reporting and minimization of sensational mistaken misrepresentations are assured

if SETI researchers follow procedures used during the Viking (and similar) missions; however some misrepresentation is inevitable given the lack of training in science of most reporters. One worrisome dimension of reportage is the likelihood that "candidate signals" – promising but unconfirmed radio bursts – will be prematurely or mistakenly reported as genuine signals from ETI's. These repeated "false alarms" can negatively affect public perceptions of science and the Microwave Survey project.

Entertainment media delve into the fantastic and do not necessarily convey accurate SETI-related imagery to the public. For example, films and novels have created a widespread public impression that interstellar travel is easy, and that physical encounters between humanity and ETI's could become commonplace. The responses of the entertainment media to an ETI detection are not easy to predict; controversy or widespread perception of "reality" following a detection might unexpectedly mute, rather than stimulate, their treatment of ETI-related subjects. Tabloid media will certainly misrepresent all stories and cannot be persuaded to do otherwise. Regular reporting of "no detection" is required, in part to forestall tabloid (and other) stories about "suppression of a discovery."

Both entertainment and news media reflect and shape the emotional backdrop of nations and the world, which in turn is likely to tilt public reactions to a detection in "anxious" or "enthusiastic" directions.

Recommendations: Educational Institutions (Paraphrased; see Chapter 5 for the full text.)

1. SETI researchers should identify ways in which information that would be useful to school teachers for presenting and dis-

playing SETI information in their classrooms can be made available to them. (This could include identification or even publication of appropriate books, which would also be recommended to librarians; see 5 below);

2. SETI researchers should make updated lists of reliable SETI books and materials available to librarians (possibly through their professional associations and networks);
3. NASA should establish a formal line of communication between the SETI community and leading professional organizations of teachers (such as the National Science Teachers Association), for routine updates about SETI and for use in the event of a detection.
4. SETI researchers should identify ways of informing the public that scientific disagreement and debate over the nature of a candidate signal is a normal and expected feature of scientific dialog;
5. Elementary school teachers should be surveyed to learn what information about SETI (and the forms in which it should be made available) would be most useful to them;
6. SETI researchers should study ways in which several dramatic events – the Sputnik launch, the discovery of Pluto, Lowell's conjectures about Mars – influenced textbooks and educational institutions.

Recommendations: News Media
(Paraphrased; see Chapter 5 for the full text.)

1. Prior to a possible detection, NASA researchers should find a timely and rou-

tine way of keeping news media (and through them, the public) informed about any candidate signals checked and discarded, or checked and still under scrutiny;

2. During the pre-detection interval, Microwave Survey news releases should regularly remind the public that dispute and debate among scientists is a normal feature of science, and that such should be expected if an ETI detection is announced;
3. NASA should devise a formal mechanism for release of information to the media, drawing heavily upon the Viking experience;
4. Studies of historic science stories that have resemblance to an ETI announcement should be conducted for insights on which communications and reports were effective, which were not, and how the public responded. (Examples are listed in the chapter.)
5. Microwave Survey researchers should devise a way of regularly reporting the absence of detected signals for the part of the search that precedes a genuine detection, and of reporting other (unanticipated) astronomical discoveries resulting from the project;
6. NASA should designate a standby team of scientists, journalists, and others who will respond to exaggerated or erroneous reports, where the latter appear to be confusing the public;
7. NASA researchers should devise a way of regularly reminding the public that "false alarms" (detections of transient radio noise or "signals" from human sources) will be a frequent and ongoing result of the HRMS research.

Recommendations: Entertainment Media
(Paraphrased; see Chapter 5 for the full text.)

1. Microwave Survey researchers should establish liaison with the entertainment industry, both to educate key media artists and producers and for staff to learn about communicating through popular forms. Briefing materials made available to news and information media should also be made available to entertainment media people. HRMS/NASA technical assistance might be made available during film and television productions of relevance to SETI.
2. HRMS researchers should consider encouraging a filmmaker to develop a major feature about SETI and receipt of an ETI signal in a responsible dramatic way.
3. Microwave Survey researchers should consider offering a non-technical seminar for producers, screenwriters, television executives and programmers to demonstrate how the HRMS works and what actual detection scenarios might be like.
4. NASA should regularly invite science fiction writers to briefings for the press and VIP's, as JPL started to do with later Voyager flyby's.
5. NASA should invite representatives from major entertainment media to serve on committees that plan the HRMS's public relations/public information efforts.
6. A summary should be compiled of scenarios to ETI detection as envisioned by works of science fiction.

Appendix

Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence

We, the institutions and individuals participating in the search for extraterrestrial intelligence,

Recognizing that the search for extraterrestrial intelligence is an integral part of space exploration and is being undertaken for peaceful purposes and for the common interest of all mankind,

Inspired by the profound significance for mankind of detecting evidence of extraterrestrial intelligence, even though the probability of detection may be low,

Recalling the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, which commits States Parties to that Treaty "to inform the Secretary General of the United Nations as well as the public and the international scientific community, to the greatest extent feasible and practicable, of the nature, conduct, locations and results" of their space exploration activities (Article XI),

Recognizing that any initial detection may be incomplete or ambiguous and thus require careful examination as well as confirmation, and that it is essential to maintain the highest standards of scientific responsibility and credibility,

Agree to observe the following principles for

disseminating information about the detection of extraterrestrial intelligence:

1. Any individual, public or private research institution, or governmental agency that believes it has detected a signal from or other evidence of extraterrestrial intelligence (the discoverer) should seek to verify that the most plausible explanation for the evidence is the existence of extraterrestrial intelligence rather than some other natural phenomenon or anthropogenic phenomenon before making any public announcement. If the evidence cannot be confirmed as indicating the existence of extraterrestrial intelligence, the discoverer may disseminate the information as appropriate to the discovery of any unknown phenomenon.
2. Prior to making a public announcement that evidence of extraterrestrial intelligence has been detected, the discoverer should promptly inform all other observers or research organizations that are parties to this declaration, so that those other parties may seek to confirm the discovery by independent observations at other sites and so that a network can be established to enable continuous monitoring of the signal or phenomenon. Parties to this declaration should not make any public announcement of this information until it is determined whether this information is or is not credible evidence of the existence of extraterrestrial intelligence. The discoverer should inform his/her or its relevant national authorities.
3. After concluding that the discovery appears to be credible evidence of extrater-

restrial intelligence, and after informing other parties to this declaration, the discoverer should inform observers throughout the world through the Central Bureau for Astronomical Telegrams of the International Astronomical Union, and should inform the Secretary General of the United Nations in accordance with Article XI of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Bodies. Because of their demonstrated interest in and expertise concerning the question of the existence of extraterrestrial intelligence, the discoverer should simultaneously inform the following international institutions of the discovery and should provide them with all pertinent data and recorded information concerning the evidence: the International Telecommunication Union, the Committee on Space Research, of the International Council of Scientific Unions, the International Astronautical Federation, the International Academy of Astronautics, the International Institute of Space Law, Commission 51 of the International Astronomical Union and Commission J of the International Radio Science Union.

4. A confirmed detection of extraterrestrial intelligence should be disseminated promptly, openly, and widely through scientific channels and public media, observing the procedures in this declaration. The discoverer should have the privilege of making the first public announcement.
5. All data necessary for confirmation of detection should be made available to the international scientific community through publications, meetings, conferences, and other appropriate means.
6. The discovery should be confirmed and

monitored and any data bearing on the evidence of extraterrestrial intelligence should be recorded and stored permanently to the greatest extent feasible and practicable, in a form that will make it available for further analysis and interpretation. These recordings should be made available to the international institutions listed above and to members of the scientific community for further objective analysis and interpretation.

7. If the evidence of detection is in the form of electromagnetic signals, the parties to this declaration should seek international agreement to protect the appropriate frequencies by exercising procedures available through the International Telecommunication Union. Immediate notice should be sent to the Secretary General of the ITU in Geneva, who may include a request to minimize transmissions on the relevant frequencies in the Weekly Circular. The Secretariat, in conjunction with advice of the Union's Administrative Council, should explore the feasibility and utility of convening an Extraordinary Administrative Radio Conference to deal with the matter, subject to the opinions of the member Administrations of the ITU.
8. No response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place. The procedures for such consultations will be the subject of a separate agreement, declaration or arrangement.
9. The SETI Committee of the International Academy of Astronautics, in coordination with Commission 51 of the International Astronomical Union, will conduct a continuing review of procedures for the detec-

tion of extraterrestrial intelligence and the subsequent handling of the data. Should credible evidence of extraterrestrial intelligence be discovered, an international committee of scientists and other experts should be established to serve as a focal point for continuing analysis of all observational evidence collected in the aftermath of the discovery, and also to provide advice on the release of information to the public. This committee should be constituted from representatives of each of the international institutions listed above and such other members as the committee may deem necessary. To facilitate the convocation of such a committee at some unknown time in the future, the SETI Committee of the International Academy of Astronautics should initiate and maintain a current list of willing representatives from each of the international institutions listed above, as well as other individuals with relevant skills, and should make that list continuously available through the Secretariat of the International Academy of Astronautics. The International Academy of Astronautics will act as the Depository for this declaration and will annually provide a current list of parties to all the parties to this declaration.

December, 1991

Regarding the Declaration

The above Declaration was developed over a period of several years by the SETI Committee of the International Academy of Astronautics, with the assistance of many experts interested in this question. In April of 1989 it was approved by the Board of Trustees of the Academy, and also by the Board of Directors of the International Institute of Space Law. Over the last three years it has been endorsed by the Committee on Space Research, by the International Astronomical Union, by the members of Commission J of the Union Radio Scientifique Internationale, and by the International Astronautical Federation.

The document is intended as a series of guidelines for individuals or organizations, national or international, engaged in carrying out radio searches for extraterrestrial intelligence. In July of 1992 it was sent by the Academy to all such individuals and organizations with a request that they give consideration to endorsing it.

In the meantime, the SETI Committee of the International Academy of Astronautics will continue to review the principles and procedures of the Declaration, and will assemble a special post-detection committee, as indicated in Principle 9 of the document. The Committee is also working on a second declaration, designed to expand the wording of Principle 8 into a process for obtaining international agreement on questions about a reply from Earth after the detection of a signal.

John Billingham
Chairman, SETI Committee
International Academy of Astronautics

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Please Note:

Included in this References section are citations corresponding to references contained previously in the text. Citations preceded by an asterisk (*) are references which are not cited in the text, but which may be useful for the interested reader. All reference citations are listed by chapter.

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